

Performance Testing

Date	03 October 2026
Team ID	FITBUDDY-TEAM
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	3 Marks

Testing Scope

The supplied FitBuddy project is primarily a local web application. Testing should therefore focus on functional response, validation, AI failure handling and database persistence rather than claiming unverified production-scale throughput.

Test Areas

Test Area	Scenario	Expected Result	Responsible Member
Registration	Valid user details	Account/profile is created	Dinesh N
Login	Valid and invalid credentials	Valid user enters application; invalid credentials are rejected	Dinesh N
Input Validation	Missing/invalid age, weight or intensity	Clear validation response	Vinesh D
Workout Generation	Valid fitness profile	Seven-day plan is generated/displayed	Deepak B
Nutrition Tip	Goal supplied	Relevant tip is displayed	Deepak B
Feedback	User submits plan feedback	Revised plan is generated and stored	Deepak B
Persistence	Reload/revisit stored information	Stored user/plan data remains available	Vinesh D
UI	Desktop and narrow viewport	Forms and results remain readable	Afraar M
AI Failure	Gemini unavailable/error	Application handles the error/fallback path without exposing the API key	Deepak B + Vinesh D

Performance Observation Method

For a formal benchmark, record request start/end time, AI response time, database operation time and total response time for repeated requests. The supplied documents do not establish a validated production throughput figure, so no unsupported numerical benchmark is claimed here.

Testing Ownership

Vinesh D coordinates test cases and evidence; Dinesh N validates backend behavior; Afraar M checks UI behavior; Deepak B checks Gemini generation and fallback behavior.